SUBJECT: DATE:

Chapter 1 (1,2,3)

- Find the direction vector of the straight line

$$\frac{x-1}{6} = \frac{y-6}{4} = \frac{z-8}{3}$$

(solution)

$(1, 6, 8)$ $d = (6, 4, 3) \rightarrow$ اتجاه الخط (direction vector)

- Find the cartesian form of the equation of the straight line passing through the points $(-7,-3,-7)$ and $(-3,-10,-4)$

(solution)

$$\vec{d} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

$$d = (-3-(-7), -10(-3), -4-(-7)) = (4,-7,3)$$

$$(-7,-3,-7), (4,-7,3)$$

$$x = -7+4t, \quad y = -3-7t, \quad z = -7+3t$$

$$t = \frac{x+7}{4}, \quad t = \frac{y+3}{-7}, \quad t = \frac{z+7}{3}$$

Cartesian form

SUBJECT: DATE:

- Give the cartesian equation of the line $\vec{r} = (-3,-2,-2) + t(4,2,4)$.

(solution)

$x = -3 + 4t$, $y = -2 + 2t$, $z = -2 + 4t$

$t = \frac{x+3}{4}$, $t = \frac{y+2}{2}$, $t = \frac{z+2}{4}$ → cartesian

chapter 2 - (a, b, c, D)

- Find an equation of the plane that satisfies the stated conditions:-

a] the plane through the origin that is parallel to the plane $4x - 2y + 7z + 12 = 0$ ↳ (0,0,0)

(solution)

$4x - 2y + 7z + 12 = 0$

vector = (4, -2, 7)

normal vector (4, -2, 7)

$4x - 2y + 7z = 0$

b] the plane that contains the line $x = -2 + 3t$, $y = 4 + 2t$, $z = 3 - t$ and is perpendicular to the plane $x - 2y + z = 5$

(solution)

$r = (-2 + 3t, 4 + 2t, 3 - t)$

$x - 2y + z = 5$

ALAQSA _ EGYPT

SUBJECT DATE

$x = -2+3t \; , \; y = 4+2t \; , \; z = 3-t$

$v = (3, 2, -1)$

$x - 2y + z = 5$

$n_1 = (1, -2, 1)$

~~Vect~~ $n_2 = v \times n_1$

$$n_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & -1 \\ +1 & 2 & +1 \end{vmatrix} = \hat{i}((2)(1) - (-1)(-2)) - \hat{j}((3)(1) - (-1)(1)) + \hat{k}((3)(-2) - (2)(1))$$

$$\cancel{n_2 = \begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} \hat{i} + \begin{vmatrix} 3 & -1 \\ 1 & 1 \end{vmatrix} \hat{j} + \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} \hat{k}}$$

$$n_2 = \cancel{(2)}\, \cancel{4\hat{i} + 4\hat{j} + 4\hat{k}}$$

$$n_2 = \hat{i}(2-2) - \hat{j}(3+1) + \hat{k}(-6-2)$$

$$n_2 = \hat{i}(0) - \hat{j}(4) + \hat{k}(-8)$$

$$n_2 = (0, -4, -8)$$

- the general equation of the plane is :-

$$0(x-(-2)) - 4(y-4) - 8(z-3) = 0$$

$$-4(y-4) - 8(z-3) = 0$$

$$-4y + 16 - 8z + 24 = 0$$

$$-4y - 8z + 40 = 0$$

$$\boxed{4y + 8z = 0}$$

SUBJECT: DATE:

7- the Plane through the Point $(-1,4,2)$ that contains the line of intersection of the Planes $4x-y+z-2=0$ and $2x+y-2z-3=0$

(solution)

$$4x-y+z-2=0$$
$$2x+y-2z-3=0$$

$$(4x-y+z-2)+\lambda(2x+y-2z-3)=0$$

بتوسيط :

$$(4+2\lambda)x+(-1+\lambda)y+(1-2\lambda)z+(-2-3\lambda)=0$$

نعوض بالنقطة $(-1,4,2)$ لإيجاد λ :

$$(4+2\lambda)(-1)+(-1+\lambda)(4)+(1-2\lambda)(2)+(-2-3\lambda)=0$$

نحسب :

$$(-4-2\lambda)+(-4+4\lambda)+(2-4\lambda)+(-2-3\lambda)$$

الثوابت : $-4-4+2-2=-8$

المجاهيل : $-2\lambda+4\lambda-4\lambda-3\lambda=-5\lambda$

اي $-8-5\lambda=0$

نعوض $\boxed{\lambda=-\frac{8}{5}}$

$4+2(-\frac{8}{5})=\frac{4}{5}$ معامل x

$-1+(-\frac{8}{5})=-\frac{13}{5}$ y

z

SUBJECT: DATE

$1-2(-\frac{8}{5}) = \frac{21}{5}$ · معامل z

الثابت : $-2-3(\frac{-8}{5}) = \frac{14}{5}$

المعادلة :

$$\frac{4}{5}x - \frac{13}{5}y + \frac{21}{5}z + \frac{14}{5} = 0$$

نضرب الكل بـ 5 :

$$4x - 13y + \cancel{12}21z + 14 = 0$$

$$\boxed{4x - 13y + 21z + 14 = 0}$$

D) the Plane through $(-1,4,-3)$ that is perpendicular to the line $x-2=t$, $y+3=2t$, $z=-t$.

(solution)

$P = (-1, 4, -3)$

$x-2=t \quad , y+3=2t \quad , z=-t$

$x=2+t \quad , y=\cancel{2t} -3+2t \quad , z=-1 \qquad d=(1,2,-1)$

$\vec{n} = (1,2,-1)$

معادلة المستوى :

$a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

$(x_0, y_0, z_0) = (-1, 4, -3)$

SUBJECT: DATE:

$(a, b, c) = (1, 2, -1)$

$1(x+1) + 2(y-4) - 1(z+3) = 0$

$(x+1) + 2y - 8 - z - 3 = 0$

$x + 2y - z - 10 = 0$

$$\boxed{x + 2y - z = 10}$$

chapter 3 (1,2,3)

1) Find the equation of the sphere of center مركز (11, 8, -5) and radius 3 in standard form.

$3 \to r$

(solution)

صيغة معادلة الكرة القياسية:

$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$

$(x-11)^2 + (y-8)^2 + (z+5)^2 = 3^2$

$$\boxed{(x-11)^2 + (y-8)^2 + (z+5)^2 = 9}$$

SUBJECT. DATE

2] Find the equation of a ~~shere~~ sphere of the center (8, −5, 10) and passing through (−14, 13, −14)

(Solutions)

$C = (8, -5, 10)$, $P = (-14, 13, -14)$

إيجاد نصف القطر :-

$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

$r = \sqrt{(-14-8)^2 + (13+5)^2 + (-14-10)^2} = 2\sqrt{346}$

معادلة الكرة :-

$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$

$(a, b, c) = (8, -5, 10)$ $r^2 = 1384$

$(x-8)^2 + (y+5)^2 + (z-10)^2 = 1384$

3] Determine the equation of a sphere with center (0, 1, 0), given that it touches one of the coordinate planes

(solution)

SUBJECT: DATE:

مركز: $(0,1,0)$

$x = 0 \quad , \quad y = 0 \quad , \quad z = 0$

المسافة

$x = 0 \quad |0| = 0 \quad , \quad y = 0 \quad |1| = 1$

$z = 0 \quad |0| = 0$

نصف القطر:

$r = 1$ لأن المسافة الوحيدة غير صفرية هي المستوى $y=0$.

الصيغة القياسية:

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

المركز: $(a,b,c) = (0,1,0)$

نصف القطر: $r = 1$

$$\boxed{x^2 + (y-1)^2 + z^2 = 1}$$